

SUMMARY

Self-taught high school software developer with 2+ years of freelance and independent project experience. Builds full-scale, real-world products across frontend, backend, and embedded/robotics systems — from AI assistants to physics simulations to competition robots.

FLAGSHIP PROJECT

The Mist Project — Founder & Lead Developer 2024 – Present

One core AI "brain" — a multi-model LLM pipeline — built out into two platforms:

Mist.AI Web (2024–2026, complete) — Full-scale AI assistant platform (Gemini, Mistral, Command R) with real-time web grounding, image understanding, and session memory. Shipped as a PWA plus Chrome/Firefox extension. ~9,000 lines of code across 10 major versions; used by real users.

Mist.AI Desktop (2025 – Present) — Downloadable Windows AI assistant (JARVIS-style) built around one continuous agent loop. Full system control — voice wake-word activation, vision-based screen understanding, real shell/PowerShell execution, filesystem automation, and browser control — with persistent memory and live captioning.

EXPERIENCE

FTC Robotics — Team Captain, Lead Programmer & Mentor 2025 – Present

Team 6183 "Triple Paradox" (DECODE season): Programmed the competition robot in Java and built/maintained the team website (React, Vite, CSS), implementing Limelight AprilTag vision for automated goal-aiming. 1st Place, Innovate Award. Elected Team Captain. Also mentor Team 21840 Kearns Robotics through 6183 outreach (1st Place, Connect Award).

Offseason: Led a 5-month rewrite of the entire robot codebase, removing a third-party framework and rebuilding every subsystem in clean, plain FTC SDK code for better team-wide reliability and readability.

Freelance Web Developer 2025 – Present

Design and build custom responsive websites for independent clients from scratch using HTML, CSS, and JavaScript — handling everything from layout and UI to deployment. Work directly with clients to scope features, gather feedback, and ship revisions, including a full contracted build (FindThisLocal.org) for an SEO Scholars program mentor.

SELECTED PROJECTS

PlasmaForge — Real-time plasma globe physics simulation. Python + Cython (custom compiled physics modules) with a Three.js frontend; true Coulomb's-law electric field and branching filament rendering. Deployed on Netlify.

SafeWord — Digital party-game web app (React/Vite PWA) with Supabase-backed card data and Firebase Firestore multiplayer room sync. Live in production.

SkillsUSA — Web Development (2025): Competed at state level building a website under timed conditions. Placed 3rd in the state.

SEO Scholars: Cohort member, 8-year college success program focused on college readiness and professional development.

National Technical Honor Society, Member

Kristian Cook

+(1) 718-674-4051
cookkristian50@gmail.com

SKILLS

Python
Java (FTC SDK)
JavaScript
React
HTML5 / CSS
Sass / Bootstrap

Tools & Platforms

Git, GitHub, VS Code
Supabase, Firebase
Fly.io, Netlify
SQLite

Certifications

CS50X — Harvard/edX (2024)
PCEP — Python Institute
Responsive Web Design — freeCodeCamp
Front End Development Libraries — freeCodeCamp

Education

Kearns Academy
High Point, NC — Class of 2028
CS, App Creation, Python Programming